



JAIPUR NATIONAL UNIVERSITY

Semester/Year End/Back Examination: May 2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Biomedical) / B.Tech. (Biotech) / B.Tech. (Chemical) / B.Tech. (Civill) / B.Tech (Mechanical) / B.Tech. (CSE) / B.Tech. (CSE-AIML) / B.Tech. (CSE-Cyber Security) / B.Tech. (Foodtech) / B.Tech. (Electrical)
Semester/Year:	2nd
Paper Code:	BTEPPCO201T22
Title of Paper:	Programming for Problem Solving
Time: 3 hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

Attempt any five Questions from section A and attempt any four Questions from section B

Question Number	SECTION-A	Marks Allotted
1.	Write a C program that reads from the user an arithmetic operator and two operands, perform the corresponding arithmetic operation on the operands using switch statement.	4
2.	Show how break and continue statements are used in a C program, with example.	4
3.	What are basic data types available in C? Write the significance of each data type.	4
4.	Write a C program to read N integers into an array A and to find the (i) sum of odd numbers, (ii) sum of even numbers, (iii) average of all numbers. Output the results computed with appropriate headings.	4
5.	Write a C program to find the area and perimeter of a circle.	4

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| 6. | What are actual parameters and formal parameters? Illustrate with example. | 4 |
| 7. | What is the difference between structure and union? Explain the C syntax of structure declaration with example. | 4 |

SECTION-B

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| 1. | What is an array? Explain the declaration and initialization of one dimensional and two dimensional array with an example. | 10 |
| 2. | Explain recursion, and write a program to find nth term of Fibonacci series. | 10 |
| 3. | Give the scope and life time of the following:
(i) External variable (ii) Static variable (iii) Automatic variable
(iv) Register variable | 10 |
| 4. | Explain with syntax and example, the different string manipulation library functions with example. | 10 |
| 5. | Answer the following.
i) What is pointer? Explain how the pointer variable declared and initialized?
ii) Write a C program to swap two numbers using call by address (pointers or reference) method. | 10 |